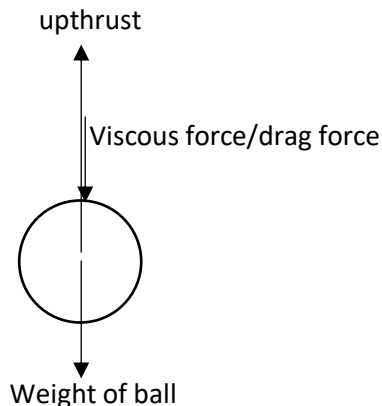


2020 Preliminary Examination H2 Physics P2 Suggested Solution

- 1(a)** Upthrust is the upward force exerted on an object in a fluid due to pressure difference between the top and bottom of the object.

(b)(i)



- (b)(ii)** Ball moving with constant speed $\Rightarrow$ no resultant force,

therefore upward forces = downward forces

$$\text{Upthrust} = \text{Drag force} + \text{weight of ball}$$

$$\text{Drag force} = \text{Upthrust} - \text{weight of ball}$$

$$\text{Drag force} = V\rho g - mg$$

$$\begin{aligned} &= \left(\frac{180 \times 10^{-3}}{800}\right)(1100)(9.81) - (180 \times 10^{-3})(9.81) \\ &= 0.66 \text{ N} \end{aligned}$$

- (c)** For ice to float, upthrust on ice must be equal to weight of ice.

After melting, the upthrust on the melted ice is the same as before, since its mass doesn't change.

Since upthrust equals weight of water displaced, the amount of displaced water is the same, before and after the ice has melted.

So the water level doesn't change.

- 2(a)** Work done is defined as the product of force and the distance moved in the direction of